

Classical (not Simulated!) Method of Moment Estimation of Life-Cycle Models

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Abstract

Structural estimation of Life-Cycle models is often performed by the Simulated Method of Moments. We show how to instead use Generalized Method of Moments estimation, without the need for simulation. Simulation is here defined as the use of random number generators, in line with the theory underlying the Simulated Method of Moments. Since simulation introduces random noise, the Generalized Method of Moments estimator is asymptotically more statistically efficient than the Simulated Method of Moments estimator. For Life-Cycle models with one endogenous state the Generalized Method of Moments estimator is also computationally faster and has smaller numerical approximation error. We illustrate by reproducing the results of Gourinchas & Parker (2002), using Generalized Method of Moments estimation where they used Simulated Method of Moments estimation.

Keywords: Structural Estimation, Generalized Method of Moments, Life-Cycle model.

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This paper can be summed up in three points.

- Generalized Method of Moments estimation is more asymptotically efficient than Simulated Method of Moments estimation.
- It is possible to solve Life-Cycle models *without* simulation.
- It is faster/more-accurate to solve Life-Cycle models *without* simulation.

These three points imply that we should estimate Life-Cycle models by the classical Generalized (not Simulated) Method of Moments — it is computationally faster, has smaller numerical approximation errors, and is more statistically efficient.

The first point, that Generalized Method of Moments is more asymptotically efficient than Simulated Method of Moments is universally known, e.g., you can find the proofs and formulae in McFadden (1989) and Pakes and Pollard (1989). The second point is long known in the Macroeconomics literature, e.g., Ríos-Rull (1995), but essentially absent from the literatures on Life-Cycle models and structural estimation, e.g., it is not mentioned at all in the guides of Low and Meghir (2017) nor Keane, Todd, and Wolpin (2011). The third point was not true a decade ago, but has become true in recent years thanks to developments in both software and hardware, especially with the Tan improvement (Tan, 2020) and the use of GPUs.

Simulated Method of Moments (SMM) was introduced by McFadden (1989). Contemporaneously a simulated minimum distance estimator was introduced by Pakes and Pollard (1989), and later a closely related simulation-based indirect inference estimator by Gourieroux, Monfort, and Renault (1993). SMM was adapted to time-series in Lee and Ingram (1991) and to Markov models in Duffie and Singleton (1993), and is used for DSGE models in Ruge-Murcia (2012). Gourinchas and Parker (2002) is an early application of SMM to Life-Cycle models, and there is a large literature with a few examples being Cagetti (2003), De Nardi, French, and Jones (2010), Kaplan (2012), and Iskhakov and Keane (2021). While Generalized Method of Moments (GMM) is well understood, we are not aware of it being previously applied to Life-Cycle models.

Implementing Simulated Method of Moments estimation for a Life-Cycle model involves an optimization routine to minimize the distance between data moments and model moments by choosing parameters θ , inside which the steps for each given value of the parameter vector θ are: (i) solve for the policy function, (ii) use the policy function to simulate panel data, (iii) evaluate moments from the panel data. Generalized Method of Moments estimation for a Life-Cycle model involves the same optimization problem, but for each given value of the parameter vector θ replace steps (ii) and (iii) with: (ii) use the policy function to iteratively compute the agent distribution and (iii) evaluate moments directly from the agent distribution. There is *no* simulation involved — *no* random number generation is used — numerical quadrature takes the place of simulation.

So for a given parameter vector θ both our GMM estimator and the SMM estimator share the same first step of solving for the policy function. Many methods to solve for the policy function

exist; e.g., Gourinchas and Parker (2002) backward-iterate on the consumption Euler equations, while our application will use value function iteration. But the choice of how to solve for the policy function is not important to us here and any of a wide variety of existing methods can be equally used.

The second step, once we have the policy function corresponding to θ , is the key difference: SMM uses the policy function together with draws from a random number generator to create shocks and simulate panel data, whereas we use numerical quadrature of the shocks and iterate on the agent distribution. The option of iterating on the agent distribution has long been known, see, e.g., Ríos-Rull (1995); however, it was little used in the 1990s and 2000s as it requires more computer memory than simulations, which was problematic on the hardware of the time. But improved computers have made iteration not just possible but faster. The recently developed Tan improvement (Tan, 2020) makes iteration possible for larger models, and means it can be orders of magnitude faster than simulation. Iterating on the agent distribution does not involve simulation, where simulation is understood as the use of a random number generator.¹

The third step is to compute the moments, from the panel data or the agent distribution. In either case this step is the simplest of the three. Panel data consists of many points, all equally weighted, and each takes a (potentially) different value. The agent distribution consists of many points, with varying weights, and each taking a (potentially) different value.² But the agent distribution typically consists of more points than the number of observations in the simulated data. This step is particularly fast when performed on a GPU, with the GPU playing an important role as its parallelization means the larger number of points when using iteration does not translate into meaningful increases in runtime. Using iteration on the agent distribution and then directly evaluating the moments is faster than simulation (for a given accuracy level) for models with one endogenous state (Ocampo and Robinson, 2023); or equivalently, iteration has smaller numerical errors than simulation, for a given runtime.

Both GMM and SMM estimation of Life-Cycle models involve approximate numerical solutions of the true model. But it is not numerical approximation errors that cause the asymptotic statistical inefficiency of SMM estimation, it is the simulation. Both GMM and SMM require an assumption that any numerical approximation errors are negligible. Kristensen and Salanie (2017) discuss the impacts of numerical approximation errors on method of moment estimators and likelihood-based

¹Two of the assumptions for SMM to be consistent and asymptotically normal are assumptions (iii) and (iv) of McFadden (1989) both of which impose conditions on the random number generation. When iterating on the agent distribution we do not use any random numbers as part of iterating the agent distribution, shocks and integrals are handled by numerical quadrature. The key to our approach is in fact hinted at in the title of McFadden (1989), 'A Method of Simulated Moments for Estimation of Discrete Response Models Without Numerical Integration'. Simulation was introduced to avoid numerical integration, while we do numerical integration by using quadrature methods.

²This description assumes a discretized agent distribution, which is both standard, and will be used in our application. Alternatives parametrizing the agent distribution are possible.

estimators, showing that they can be understood as introducing bias into the estimators.³ The GMM estimator, viewed as an algorithm, represents a better speed-accuracy trade-off frontier than SMM. This means that in practice GMM will be faster to compute and have smaller numerical approximation errors than SMM. GMM’s advantage of having smaller numerical errors (for any given runtime) is in addition to GMM being more statistically efficient than SMM.

Our GMM estimator does have one limitation versus SMM, namely that the quadrature is more susceptible to the curse-of-dimensionality than the simulation underlying SMM. In practice this is typically not an issue, since most approaches to solving for the policy function involve using quadrature to discretize the shocks, and hence if we can solve the policy function then we will not run into the curse-of-dimensionality when iterating the agent distribution and calculating the moments. That is, if we are able to estimate the model full-stop, then we can estimate it by GMM. Sometimes though, models with a large number of shocks are used, together with a heavy approximation method for solving the policy function like the EMax operator (Keane and Wolpin, 1994), and in these cases SMM will remain the only option. But this does not apply, e.g., for the models in any of the SMM estimation of life-cycle papers mentioned earlier.⁴ In summary, for life-cycle models GMM estimation is to be preferred to SMM estimation wherever it is feasible, which is roughly any life-cycle model that does not include a large number of exogenous shocks.

We first explain the general setting of method of moment estimation of a Life-Cycle model, and how implementing the estimation differs with and without simulation. We then describe existing results on asymptotic statistical efficiency, as well as the use of efficient GMM. Computer runtimes are reported showing that the GMM estimator is faster than the SMM estimator. These runtimes are based on the model of Gourinchas and Parker (2002). Gourinchas and Parker (2002) estimate a life-cycle model by SMM, and the last part of this paper explains their model and target moments and then re-performs their estimation using GMM to illustrate how GMM can be used in place of SMM to estimate life-cycle models.

³Kristensen and Salanie (2017) characterize both non-stochastic approximations (like numerical approximation error) which introduce bias, and stochastic approximations (like simulation error) which both introduce bias and reduce statistical efficiency. In their setting, our main point can be understood as saying that for life-cycle models GMM has bias due to non-stochastic numerical approximation error, while SMM has a larger bias (for a given runtime) from a combination of non-stochastic numerical approximation error and stochastic simulation approximation and on top of this a loss of statistical efficiency from stochastic simulation approximation. Numerical approximation error was largely ignored in the original simulation-estimators literature: Duffie and Singleton (1993) mention the issue of numerical error in passing in footnote 2 on page 932 stating that ‘At a minimum, the methods described in this paper apply to the approximate model if approximations are used to solve [the model]. They may apply to the original model if the approximation error can be made negligible as the sample size increases’. Numerical error is entirely omitted in McFadden (1989), Pakes and Pollard (1989) and Lee and Ingram (1991).

⁴The use of a large number of shocks is more common in maximum likelihood or Bayesian likelihood estimation as there ‘enough’ shocks must be included to avoid the likelihood being singular.

1 Estimation of Life-Cycle Models

We are interested in J -period Life-Cycle models that involve solving dynamic programs like,

$$V_j(a, z, e) = \max_{d, a'} F(d, a', a, z, e; \theta) + \beta E[V_{j+1}(a', z', e')|z], \quad j = 1, 2, \dots, J$$

where d is a decision variable, a is an endogenous state, z is a Markov exogenous state with transition matrix Π_z , e is an i.i.d. exogenous state with distribution Π_e , β is a discount factor, and $F()$ is a return function (typically, e.g., a combination of a utility function and budget constraints). We use $'$ to denote the next period. It is assumed there is some terminal V_{J+1} , and that agents begin period 1 in some state (e.g., drawn from an initial agent distribution). Solving finite-horizon Life-Cycle models to compute the optimal policy function is standard and so we will not describe it here.⁵ Once the policy function has been solved we iterate the model and compute model moments.

Let θ be the model parameters to be estimated and M_m be some model moments. Then we can consider the model as a function from θ to M_m , which we denote by $M_m(\theta)$.⁶

We want to estimate the model parameters θ by targeting a vector of moments of the observed data, M_d . We need to assign a matrix of weights, W , that gives the relative importance of each of these moments. In practice, often only a subset of the model parameters are estimated, the rest might be calibrated in a step before estimation.⁷

Method of moments estimation aims to minimize the difference between the data moments and model moments by solving

$$\theta^* = \arg \min_{\theta} [M_d - M_m(\theta)]' W [M_d - M_m(\theta)] \quad (1)$$

Simulated method of moments replaces the model moments M_m , with simulated model moments $M_{m,s}$, and involves solving

$$\theta^* = \arg \min_{\theta} [M_d - M_{m,s}(\theta)]' W [M_d - M_{m,s}(\theta)]$$

Readers familiar with GMM will notice that this is not a standard formulation. Our definition only applies when the moment conditions are 'separable', which is the case in essentially all applications

⁵The results of this paper can be applied to a far wider range of Life-Cycle models such as making all parameters and shocks age-dependent, adding semi-exogenous shocks, or alternative preferences like Epstein-Zin or Gul-Pesendorfer, or to quasi-hyperbolic discounting. Or to models in which the next period endogenous state a' cannot be directly chosen, such as portfolio-choice models and models with uncertain human-capital accumulation. They apply when the states and decision variables are of any finite-dimension. They also apply to models with different permanent types of agents. We stick to a standard plain-vanilla Life-Cycle model setup in this paper—close to the one we will estimate—for expositional purposes.

⁶A formal proof providing some conditions under which a standard Life-Cycle model is a function from model parameters θ to model moments can be found, e.g., as part of Kirkby (2019).

⁷ θ may include parameters that determine the exogenous shock processes, or the initial agent distribution, and may include the discount factor. Our notation omits these (potential) dependencies to make it easier to read.

of GMM estimation to Life-Cycle models. The next paragraph provides the standard definition of GMM and shows how when the moment conditions are separable we get the above formulation.

Equation (1) is GMM, but does not make explicit the requirement that M_d is a moment, as well as only applying when the moment conditions are separable. For GMM, our moment conditions have to be population averages that evaluate to zero (which are implemented using the sample averages). Our formal moments conditions are $M_d - M_m = 0$, and not about M_d nor M_m separately. Method of moments requires the moment condition to be defined as $\frac{1}{N} \sum_{i=1}^N f(x_i, \theta) = 0$, that is as an average of a function of individual observations being equal to zero, where $\{x_i\}_{i=1}^N$ are the observed data, and the function $f(x_i, \theta)$ is defined as a (vector-valued) function of a single observation. Since M_d is a data moment, there must be a function $m()$ such that $M_d = \frac{1}{N} \sum_{i=1}^N m(x_i)$. So we can rewrite $M_d - M_m(\theta) = \frac{1}{N} \sum_{i=1}^N m(x_i) - M_m(\theta) = \frac{1}{N} \sum_{i=1}^N (m(x_i) - M_m(\theta))$, and since $(m(x_i) - M_m(\theta))$ is a function of a single observation x_i , we have that our formal moment condition is $\frac{1}{N} \sum_{i=1}^N (m(x_i) - M_m(\theta)) = 0$. Defining $g(\{x_i\}_{i=1}^N, \theta) = \frac{1}{N} \sum_{i=1}^N (m(x_i) - M_m(\theta))$, we can write the more formally precise definition of our GMM estimator as $\theta^* = \arg \min_{\theta} g(\{x_i\}_{i=1}^N, \theta)' W g(\{x_i\}_{i=1}^N, \theta)$.⁸ We use equation (1) as it is closer to both how estimating Life-Cycle models is commonly described, and to how the estimator is coded.

Note that for most GMM estimators it is not possible to rewrite the moment conditions as two separate terms, one which depends on the data, here $M_d = \frac{1}{N} \sum_{i=1}^N m(x_i)$, and one which is a constant (given θ), here $M_m(\theta)$. Notice that the first term is a random variable and the second term is just a constant, which will be important later when discussing efficient GMM. That the moment conditions are separable is not necessarily true when performing GMM estimation a Life-Cycle model but in practice it almost always is, including in all of the five papers we cited as examples of SMM estimation of Life-Cycle models in the Introduction.⁹

Regardless of whether doing GMM or SMM we have the same pseudo-code for the main optimization problem, as described in Algorithm 1,

Algorithm 1 Simulated/Generalized Method of Moments Estimation: Optimization

- 1: Define M_d and W . ▷ Estimate them from observed data.
 - 2: Guess initial θ .
 - 3: Use optimization to solve $\min_{\theta} [M_d - M_{model}(\theta)]' W [M_d - M_{model}(\theta)]$
-

There are a variety of optimization algorithms available to implement this and we will not discuss them here.¹⁰ The difference between Simulated Method of Moments and Generalized Method of

⁸The language of the formal 'moment conditions' being made up a moment of the data and a moment of the model is clunky but standard. We are unfortunately not aware of a nicer terminology for describing this issue.

⁹Separability is better understood as a property of the moments, rather than the model, as some examples in simple (not Life-Cycle) models at github.com/robertdkirby/GMMSeparableMoments illustrate. Note that it is still possible to GMM estimate Life-Cycle models in which the moment conditions are not separable, but separability allow us to more intuitively write the description of the problem to be estimated, and to simplify both the code to implement the estimation, and the implementation of efficient GMM.

¹⁰Our codes use CMA-ES (Covariance Matrix Adaptation-Evolutionary Strategy, which is slow but robust. An

Moments is in how we compute $M_{model}(\theta)$.¹¹

SMM uses the pseudo-code in Algorithm 2 to calculate M_{model} ,

Algorithm 2 Simulated Method of Moments Estimation: $M_{model}(\theta)$

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1: Given  $\theta$ .
2: Solve Life-Cycle model to get policy  $g_j(a, z, e; \theta)$ .
3: for  $ii = 1 : sN$  do
4:   Simulate a  $J$  period life cycle
5:   Draw initial agent state  $(a_1, z_1, e_1)$  from initial agent distribution.
6:   Store  $x_{i,1} = [a_1, z_1, e_1]$ 
7:   for  $j = 2 : J$  do
8:     Draw (random)  $e_j$  from  $\Pi_e(\cdot)$ 
9:     Draw (random)  $z_j$  from  $\Pi_z(\cdot|z_{j-1})$ 
10:     $a_j = g_{j-1}(a_{j-1}, z_{j-1}, e_{j-1}; \theta)$ 
11:    Store  $x_{i,j} = [a_j, z_j, e_j]$ 
12:   end for
13:   Store  $x_i = [x_{i,1}, x_{i,2}, \dots, x_{i,J}]$ .
14: end for
15: Calculate moments  $M_{model}(\theta) = \frac{1}{sN} \sum_{i=1}^{sN} f(x_i)$ 
16: return  $M_{model}(\theta)$ 

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where N is the number of observations used to calculate the data moments M_d , and sN is the number of observations used to calculate the simulated moments M_d , with s a positive integer.¹² There is a random number generator used in this process for drawing e_j and z_j .¹³

By contrast, GMM uses the pseudo-code in Algorithm 3 to calculate M_{model} , where $\#\mu$ is the number of grid points in μ (which follows from the number of points in the a , z and e grids across all ages). Notice that there are *no* random draws here! We use quadrature to implement numerical integration instead of making random draws.

A few comments. (i) Notice that solving the policy function involves evaluating numerical integrals (as we have the *expection* of next period value function), and we assume that this is done by numerical quadrature as is standard in practice. (ii) The theory of SMM requires the same sequence from a random number generator to evaluate the model moments across different values of θ .¹⁴ (iii) When coding the step $\mu_{j+1} = T'_j \mu_j$ of the GMM estimator the Tan improvement should be used to increase speed (by breaking this into two separate substeps, Tan (2020)). (iv)

alternative to using optimization methods is to transform the problem to a Laplace-type estimator, a setting where MCMC is possible (Chernozhukov and Hong, 2003).

¹¹The notation $M_{model}(\theta)$ corresponds to either the $M_m(\theta)$ or the $M_{m,s}(\theta)$ from earlier, depending on whether we are doing GMM or SMM.

¹²In our application all the target moments are age-conditional moments, and this code corresponds to calculating each data moment from N observations, and each simulated moment from sN observations. If our target was, e.g., total assets of all households, then it would be calculated from NJ data observations and sNJ simulated observations.

¹³A random number generator will typically also be required to draw from the initial agent distribution.

¹⁴Modern algorithms for stochastic gradient descent could perhaps allow for this assumption to be relaxed but at the cost of introducing further noise into the estimator.

Algorithm 3 Generalized Method of Moments Estimation: $M_{model}(\theta)$

- 1: Given θ .
 - 2: Solve Life-Cycle model to get policy $g_j(a, z, e; \theta)$.
 - 3: Define grid on endogenous state a_grid
 - 4: Use quadrature to discretize shocks: $[z_grid, Q_z] = discretize(\Pi_z)$ and $[e_grid, Q_e] = discretize(\Pi_e)$.
 - 5: Set period one distribution, μ_1 , equal to the initial agent distribution on the (a, z, e) grids.
 - 6: **for** $j = 1 : J - 1$ **do**
 - 7: Define $T_j = g_j(\cdot, \cdot, \cdot; \theta) \circ Q_z \circ Q_e$. $\triangleright T_j$ is a Markov transition matrix on (a, z, e)
 - 8: $\mu_{j+1} = T_j' \mu_j$
 - 9: **end for**
 - 10: Store $\mu = [\mu_1, \mu_2, \dots, \mu_J]$.
 - 11: Calculate moments $M_{model}(\theta) = \sum_{i=1}^{\# \mu} f(a, z, e, j) d\mu(a, z, e, j)$
 - 12: **return** $M_{model}(\theta)$
-

Depending on the details of the application, things like the quadrature step in the GMM will be independent of θ and so can be precomputed once rather than being done for each θ . (v) Depending on the algorithm used to get the policy, the grids and quadrature might be completed prior to the calculation of the policy.

2 Asymptotic Statistical Efficiency

We report existing results on the relative asymptotic inefficiency of SMM relative to GMM. We will follow Pakes and Pollard (1989) who report results for Simulated Minimum Distance Estimators (of which SMM is an instance) in *i.i.d.* data. While results also exist for time-series data we are not aware of any extension to panel data¹⁵ for SMM estimators, so we report the *i.i.d.* results.

For most Life-Cycle models, both GMM and SMM estimators are consistent. SMM requires the same assumptions as GMM, plus additional assumptions that the simulated moments are consistent for the true model moments. For most Life-Cycle models, both GMM and SMM estimators are asymptotically normal. SMM requires the same assumptions as generalized, plus additional assumptions that the simulated moments are uniformly 'close' in probability to the true moments.

Let N be the number of observations used to calculate the data moments. Assume that the simulated model moments are calculated based on sN observations, where s is a positive integer. Both McFadden (1989) and Pakes and Pollard (1989) base their asymptotic results on this setup of N and sN observations, with $N \rightarrow \infty$. Let θ_0 be the true value of the parameter to be estimated.

If the assumptions required for GMM to be consistent and asymptotically normal are satisfied, then we get,

$$\sqrt{N}(\hat{\theta} - \theta_0) \rightarrow N(0, \Sigma) \tag{2}$$

¹⁵With asymptotics that have a fixed time horizon while the cross-sectional dimension goes to infinity.

where $\Sigma = (J'WJ)^{-1}J'W\Omega WJ(J'WJ)^{-1}$, W is the weighting matrix on the moments, J is the Jacobian matrix of derivatives of model moments with respect to θ ($J \equiv \frac{\partial M_m}{\partial \theta}|_{\theta=\theta_0}$), and Ω is the covariance matrix of the moment conditions.^{16,17}

If the assumptions required for SMM to be consistent and asymptotically normal are satisfied, then we get,

$$\sqrt{N}(\hat{\theta} - \theta_0) \rightarrow N(0, (1 + 1/s)\Sigma) \quad (3)$$

where Σ is the same as for Generalized Method of Moments.

Notice that the only difference between (2) and (3) is that the expression for the variance of the SMM is $1 + 1/s$ times as large as for Generalized Method of Moments. When there are N observations in the data, we are using sN observations for the simulations. So whereas it is possible to set very large s to reduce the loss of asymptotic efficiency, the higher we set s the longer the simulations will take. For example, if we set $s = 10$, then the asymptotic variance of SMM will be only 1.1 times larger, whereas if we set $s = 2$ the asymptotic variance will be 1.5 times larger than for Generalized Method of Moments. Intuitively, the simulations introduce an additional degree of randomness to the estimator, and this additional randomness scales inversely with s (as there are s simulated observations for each data observation).¹⁸

This establishes the asymptotic statistical inefficiency of SMM, relative to GMM, which as previously stated is well-established and well-known in the literature. The rest of this section discusses further aspects of the estimation.

We can perform these estimators with any (positive definite) weighting matrix W . The most asymptotically efficient estimator sets $W = \Omega^{-1}$, that is to use the inverse of the covariance matrix of the data moments as a weighting matrix. In this case the asymptotic variance for Generalized Method Moments simplifies to $(J'\Omega^{-1}J)^{-1}$ and for Simulated Method of Moments to $(1+1/s)J'\Omega^{-1}J$. So again, Simulated Method of Moments remains less efficient, with an asymptotic variance $1 + 1/s$ times as large.¹⁹ Ω is unknown so actually implementing this estimator will involve using the empirical estimate $\hat{\Omega}$, in place of Ω , so $W = \hat{\Omega}^{-1}$.

It turns out that for the kinds of moment conditions used in the estimation of Life-Cycle models, efficient GMM is easy to implement as $\hat{\Omega}$ simplifies to being the covariance matrix of the data moments and so can be calculated directly from the data without using the model. The key is the

¹⁶In principle, J is the Jacobian matrix of the moment conditions for GMM, but as long as the moments are separable, since the data moments are independent of θ , this simplifies to just the Jacobian of the model moments.

¹⁷In the just-identified case, where the number of moment conditions equals the number of parameters to be estimated, we get $\Sigma = (J'\Omega^{-1}J)^{-1}$

¹⁸That SMM has $(1 + 1/s)$ times the asymptotic variance of Generalized Method of Moments is largely independent of the setting and appears in the results of all of McFadden (1989), Pakes and Pollard (1989), Lee and Ingram (1991) and Duffie and Singleton (1993).

¹⁹McFadden (1989), Pakes and Pollard (1989), and Lee and Ingram (1991) all provide asymptotic variance results with both a generic W , and with $W = \Omega^{-1}$. Duffie and Singleton (1993) only provide results for $W = \Omega^{-1}$.

observation, made earlier, that our moment conditions are separable as $\frac{1}{N} \sum_{i=1}^N (m(x_i) - M_m(\theta)) = \frac{1}{N} \sum_{i=1}^N m(x_i) - M_m(\theta) = M_d - M_m(\theta)$, where the first term depends on the observed data and the second term is a constant. Therefore, Ω , which is the covariance matrix of our estimation moments $M_d - M_m(\theta)$, simplifies to just the covariance matrix of M_d , as $M_m(\theta)$ is a deterministic constant— $Cov(Z + c) = Cov(Z)$ for any random variable Z and constant c . So we can estimate $\hat{\Omega}$ as the covariance-matrix of our data moments, without any role for our model $M_m(\theta)$, and use (the inverse of) this estimate for our weighting matrix for efficient GMM. This provides an important contrast to SMM, as efficient SMM uses the covariance matrix of the moment conditions $M_d - M_{m,s}(\theta)$, but now $M_{m,s}(\theta)$ is a random variable (as it is simulated), and so we cannot calculate this from the data.

Instead two-iteration efficient SMM must be used. The first-iteration uses any weighting matrix, such as the identity matrix, to get a consistent estimate of θ , which we will call θ_1 . The second-iteration involves calculating the covariance matrix of the moment conditions $M_d - M_{m,s}(\theta_1)$, so based on θ_1 , and then using the inverse of this as the weighting matrix in the second-iteration. The second-iteration is required in SMM but not GMM as we have to account for the variance of the simulation and the covariance between the simulations and the data. Efficient estimation is substantially different for GMM and SMM estimation of Life-Cycle models, as the separability of the moment conditions means efficient GMM is done in a single step, whereas efficient SMM requires two iterations.^{20,21} So when efficient estimation is of interest—three of the five examples of SMM estimation of Life-Cycle papers mentioned in our Introduction implement the efficient estimator—the runtime gains from GMM over SMM can be substantial as they avoid the need for the second iteration, as well as substantially simplifying the estimation of $\hat{\Omega}$.

$W = \hat{\Omega}^{-1}$ is asymptotically efficient but it can be problematic in small samples (just as weighted least squares is more asymptotically efficient than OLS but typically performs worse in small samples). Altonji and Segal (1995) explain why $W = \hat{\Omega}^{-1}$ can be problematic in small samples: because the weighting matrix $W = \hat{\Omega}^{-1}$ may be correlated with the estimated moments, $\hat{M}_d - M(\theta)$, and this causes the estimator to be biased. They also provide intuition for when this is likely to happen, namely when our target moments include both first and second moments (rather than, e.g., just first moments as in our application to the model of Gourinchas and Parker (2002)).²² When

²⁰In most situations the moment conditions for GMM are not separable in this manner, so efficient GMM cannot be done in a single step and two-iteration efficient GMM is needed. Life-cycle models are unusual, but not unique, in typically using separable moment conditions for estimation. All five of the example papers on SMM estimation of Life-Cycle models cited in the Introduction used moment conditions that are separable. This is better understood as a property of the moment conditions, rather than of the model, as illustrated by the examples at github.com/robertdkirkby/GMMSeparableMoments. It would be possible to use non-separable moments in the estimation of a Life-Cycle model, but is exceedingly rare in practice.

²¹Separable moment conditions mean that we can estimate the covariance matrix of the moment conditions without a value of θ , as it is independent of θ . But notice that we **cannot** evaluate the moment conditions themselves without a value of θ .

²²For example, consider a setting where we estimate two moments of data $\{x_i\}_{i=1}^N$, the first is the mean of x , and the second is a variance of x . Call θ the parameter vector to be estimated, and let $g_1(\theta)$ be the mean for x of the model, $g_2(\theta)$ be the variance for x of the model. So our two moments for GMM are calculated as $\frac{1}{N} \sum_{i=1}^N (x_i - g_1(\theta))$

using $W = \hat{\Omega}^{-1}$ is likely problematic (e.g., when we target both first and second-order moments), efficient GMM should be avoided; a popular alternative is to use $W = \text{diag}(\hat{\Omega})^{-1}$ (a matrix with zeros on the off-diagonals, and with the diagonal being the elementwise-inverse of the diagonal of Ω), although it is not clear this avoids the bias.²³

To evaluate the asymptotic variance, Σ , we need J , W and Ω . W , and J can be easily computed once we have estimated the parameter vector,²⁴ that leaves Ω . So we need an empirical estimate $\hat{\Omega}$ for the covariance of the moment conditions even if we do not use it's inverse as a weighting matrix.

Results for GMM and SMM with time-series data are similar except we use T instead of N , and Ω (the covariance matrix of the moment conditions) is more complicated due to the presence of autocorrelation, and must be estimated with the Newey-West estimator or similar.

We have seen that SMM is always asymptotically inefficient relative to GMM. In the next section we will see that for many Life-Cycle models it also takes longer to compute.

3 Runtime

Both SMM and GMM estimation for a Life-Cycle model share many steps. Both solve for the policy function, both choose the parameter vector to minimize the (weighted) distance between the data moments and the model moments. So the only differences in runtimes will come from the difference between simulating panel data and calculating the moments from it, versus iterating the agent distribution and calculating the moments from it. Iterating the agent distribution is faster. We report runtimes relating to the model of Gourinchas and Parker (2002), described in the next section, to illustrate.

Simulations used in the SMM estimator require sN observations, where N is the number of data observations and s is a positive integer. Gourinchas and Parker (2002) calculate each moment from

and $\frac{1}{N} \sum_{i=1}^N ((x_i - \bar{x})^2 - g_2(\theta))$. Consider the variance-covariance matrix of the moments, the inverse of which we would like to use as the weighting moment in efficient GMM. The first entry of the variance-covariance matrix is the 'square' of our first moment, namely $\frac{1}{N} \sum_{i=1}^N (x_i - g_1(\theta))^2$. This is similar to our second moment, and the two will typically be correlated in small samples. So we have that one of our moments, is correlated with an element of our weighting matrix, which is the inverse of this variance-covariance matrix. This is likely to occur in any situation where we use the first and second moments as targets, since the variance-covariance matrix element for the variance of the first moment is likely to correlate with our second moment targets. The examples in Altonji and Segal (1995) are about estimating covariance structures using GMM.

²³Altonji and Segal (1995) advocate using $W = \mathbb{I}$ in small samples, but this means the parameter estimate is scale-dependent (e.g., different parameter estimates depending on whether we measure a moment in dollars or thousands of dollars, as the weights do not reflect the units). Using a W which is diagonal and avoids scale dependence can be better. Note that $W = \text{diag}(\hat{\Omega})^{-1}$ is conceptually subject to the same bias from the correlation between data moments and weighting matrix as causes small sample issues with $W = \hat{\Omega}^{-1}$.

²⁴Recall that J is the Jacobian matrix of derivatives of the model moments with respect to the parameter vector, evaluated at the estimated parameter vector. This can be computed by taking finite differences of the model moments for small-epsilon changes in the parameter vector, or using automatic differentiation.

roughly 1000 observations,²⁵ So the minimum possible value of sN is 1000 ($s = 1$ with $N = 1000$). We use this minimum for the following runtimes so that the simulation is as fast as possible. Note that $s = 1$ is not often used in practice as it leads to a substantial increase in the standard deviation of the estimated parameters.

Computing the agent distribution via iteration takes 0.29 seconds, and 0.07 seconds are required to calculate the age-conditional means (of consumption and income) from this, for a total of 0.36 seconds. Simulating panel data with 1000 individuals takes 0.51 seconds, and 0.02 seconds are required to calculate the age-conditional means from this, for a total of 0.52 seconds.²⁶ Omitting the time required to compute the policy function which is identical across Generalized and Simulated Method of Moments, we have that iteration takes just 69% of the runtime required for simulation. The most time consuming step is computing the policy function, which takes 4.06 seconds in this example, so the speed gains of GMM are here modest. But since GMM already has higher asymptotic efficiency, the fact that it is also faster to solve means that there is no reason to use Simulated Method of Moments in simple Life-Cycle models.

Whereas the runtime gains of iterating the agent distribution were modest, this was using simulations with $s = 1$. Even with only $s = 1$ the iteration is faster! If a more common number like $s = 10$ was used for simulation the runtime advantages become substantial (simulation runtimes are essentially linear in the number of simulated observations, so they would be 10 roughly ten times as slow). For context, Gourinchas and Parker (2002) used $s = 20$ (20,000 simulated individuals), and Jorgensen (2023) used $s = 500$ (500,000 simulated individuals) when estimating this same model.²⁷ If we set $s = 20$, as used by Gourinchas and Parker (2002), then evaluating a value of θ would take 4.42 seconds for GMM, including the value function iteration, compared to 14.46 seconds for SMM, so GMM is now substantially faster.

The actual ratio of runtimes will depend on hardware. Panel data simulations will be easily parallelized across CPUs and so runtimes drop (inverse) linearly with the number of CPUs. We report runtimes from a desktop with hardware that is reasonably balanced between CPU and GPU, in the sense the neither costs notably more than the other (6 CPUs and a GPU with 8GB of GDDR memory). A definitive comparison of runtimes is not attempted since differences not only impact run times but also the accuracy of moments, and both runtimes and accuracy will differ by the type of moment (mean, median, etc.). Ocampo and Robinson (2023) find that iteration combines

²⁵1000 is the roughly the number of data observations (per age) in Gourinchas and Parker (2002). They report 40,000 households (their page 66), but each household is only observed for one year and this is (not evenly) divided by the 40 model periods.

²⁶All runtimes reported as the average across 100 runs. All these runtimes include overhead from VFI Toolkit. The calculation of the statistics from the panel data is faster than the agent distribution. There are two reasons for this. The first is that the panel data is just 40,000 observations (1000 individuals, by 40 ages) whereas the agent distribution has ~8million points. The second is that evaluating consumption as a function of the state-space is done as part of computing means from the agent distribution, but as part of simulating the panel data, and so appears in different components of runtimes across the two examples.

²⁷Page 60 of Gourinchas and Parker (2002) reports the 20,000. Page Line 165 of Jorgensen (2023) codes [solve_GP2002.py](#) sets `simN = 500000`, lines 237-244 make clear that this is the number of individual simulations.

more accurate moments with lower runtimes, relative to simulation, for longitudinal moments. Our experience is that agent distribution iteration provides a better speed/accuracy trade-off across essentially all one endogenous state Life-Cycle models and all moments. It is true for the variety of Life-Cycle models in the ['Intro to Life-Cycle Models'](#).

Note also that in situations where efficient GMM is of interest, the runtime gains for GMM over SMM are substantially increased. For GMM estimation of Life-Cycle models we can simply estimate $\hat{\Omega}$ from the observable data, whereas for SMM estimation of Life-Cycle models this is not possible and we must do two-iteration efficient SMM. So when we want the efficient GMM estimator, we can avoid the second-iteration which SMM would require. The runtime gain from not requiring two iterations is substantial, but not as high as fifty-percent as the first-iteration parameter estimate provides a good initial guess for the second-iteration and so the second iteration is much faster than the first.

4 Application to Gourinchas & Parker (2002)

We now use GMM to estimate the Life-Cycle model of Gourinchas and Parker (2002). Our estimates serve as a proof-of-concept. We use the *EstimateLifeCycleModel_MethodOfMoments()* command from VFI Toolkit, and the whole estimation requires less than 10 additional lines of code beyond those to setup and solve the model. The model was developed by Carroll (1997) and then estimated by Gourinchas and Parker (2002), and has been reestimated in the contexts of both understanding the sensitivity of parameter estimates to the estimation moments (Andrews, Gentzkow, and Shapiro, 2017) and of understanding the sensitivity of parameter estimates to the calibrated parameters (Jorgensen, 2023).

Gourinchas and Parker (2002) uses SMM to estimate the $J = 40$ period Life-Cycle model,

$$\begin{aligned} V_j(a, z, e) &= \max_{a'} v_j \frac{c^{1-\rho}}{1-\rho} + \beta E[V_{j+1}(a', z', e')|z] \\ \text{s.t. } c + a' &= Ra + ze \\ a' &\geq 0 \end{aligned}$$

where a is assets, c is consumption, v_j is a utility-shifter that depends on age j . Earnings follows an exogenous stochastic process comprised of a Markov z and an i.i.d. e . The terminal period value function is given by $V_{J+1} = \kappa v_{J+1} \frac{(X_{J+1} + H_{J+1})^{1-\rho_J}}{1-\rho_J}$, where $X_{J+1} = Ra_{J+1} + z_{J+1}$ and $H_{J+1} = h z_{J+1}$; by assumption $z_{J+1} = z_J$.^{28,29}

²⁸Gourinchas and Parker (2002) do not explicitly include $a' \geq 0$ in their setup. But they follow Carroll (1997) in combining an assumption that agents must die with non-negative assets (that $a_{J+1} \geq 0$) with an income process that has a positive probability of taking a zero value in each period. Together these two assumptions imply that $a' \geq 0$ in every period.

²⁹We make one change from the model estimated by Gourinchas and Parker (2002), namely they impose that $\rho_J = \rho$, while we allow them to differ. We discuss why this was done below when describing our estimates.

The (non-stationary) Markov shocks follow a non-stationary unit-root with drift,

$$\begin{aligned} \log(z_j) &= g_j + \log(z_{j-1}) + n \\ n &\sim N(0, \sigma_{z,n}) \end{aligned}$$

The i.i.d shocks are assumed to follow

$$e \begin{cases} \text{with probability } p : & e = 0 \\ \text{with probability } 1 - p : & u = \log(e), u \sim N(0, \sigma_u) \end{cases}$$

Agents in period $j = 1$ are distributed over assets as a log-normal distribution $\log(a) \sim N(\omega_{26,mean}, \omega_{26,stddev}^2)$. The article of Gourinchas and Parker (2002) does not discuss how the initial values of z and e are set. The code of Jorgensen (2023) sets period 0 to the mean value of $\log(Y_1)$, and then does one draw of the permanent shock innovations to get the period 1 permanent shock distribution; and uses the standard transitory shock distribution. The agent distribution is set to the cross-product of the asset, permanent shock, and transitory shock distributions.

Some parameters are calibrated, specifically g_j , $\sigma_{z,n}$, σ_u , $\omega_{26,mean}$, and $\omega_{26,stddev}$. We set all the parameters to their values from Gourinchas and Parker (2002) and report them in the Appendix.

The parameters β , ρ , h , κ and ρ_J are estimated by Generalized Method of Moments (by SMM in Gourinchas and Parker (2002)).

The target data moments are the log of the age-conditional mean of consumption (so $J = 40$ targets). We perform two estimates, following Gourinchas and Parker (2002). First, we do a robust estimate using $W = I$, i.e., the weighting matrix is the identity matrix. Second, we do an efficient estimate using $W = \hat{\Omega}^{-1}$, i.e., the weighting matrix has the inverse of the empirical variance of each moment on the diagonal, and zeros on the off-diagonal.³⁰ Note that we can implement the efficient GMM directly, whereas Gourinchas and Parker (2002) had to use two-iteration efficient SMM.

Our estimates are reported in Table 4. Note that our results differ slightly from those of Gourinchas and Parker (2002) who estimate $\beta = 0.960$ and $\rho = 0.514$, and implicitly $h = \gamma_0/\gamma_1 = 0.0015/0.071 = 0.02111$.³¹

Our estimates differ because while they lay out precisely the model described above (on their pages 51-54) they do not estimate this exact model. Instead they convert the model into consumption Euler equations and solve these to get the policy function, instead of solving by value

³⁰We take the empirical moments and their variances from the GitHub repo of Jorgensen (2023), who in turn got them from the replication materials of Andrews, Gentzkow, and Shapiro (2017). Note that this is the inverse of the variance-covariance matrix of the data moments; the zeros on the off-diagonal occur because the underlying dataset was a repeated cross-section and they are all age-conditional moments.

³¹Andrews, Gentzkow, and Shapiro (2017) got $\beta = 0.9574$ and $\rho = 0.6526$.

Estimated Parameters			
Robust estimates ($W = \mathbb{I}$)			
	<i>Point Estimate</i>	<i>90% Conf. Int</i>	<i>Description</i>
β	0.961	[0.952, 0.969]	discount factor
ρ	0.294	[0.053, 0.535]	curvature of utility fn
h	10.939	[-13.615, 35.494]	role of permanent income in warm-glow
κ	3.029	[0.258, 5.801]	relative importance of warm-glow
ρ_J	0.306	[0.138, 0.473]	curvature of warm-glow utility fn
Efficient estimates ($W = \Omega^{-1}$)			
	<i>Point Estimate</i>	<i>90% Conf. Int</i>	<i>Description</i>
β	0.961	[0.952, 0.969]	discount factor
ρ	0.297	[0.058, 0.535]	curvature of utility fn
h	37.459	[-37.781, 112.699]	role of permanent income in warm-glow
κ	2.927	[0.111, 5.742]	relative importance of warm-glow
ρ_J	0.286	[0.135, 0.438]	curvature of warm-glow utility fn

Table 1: GMM estimates for Gourinchas & Parker (2002)

function iteration. While the switch from value function to consumption Euler equations does not change the solution, they do make a change to period $J + 1$ as part of the switch. When setting up the value function they parametrize V_{J+1} by κ and h , as above (but assuming $\rho_J = \rho$). For the consumption Euler equations they instead parametrize c_{J+1} by γ_0 and γ_1 .³² Their paper provides the form for V_{J+1} that we use here (with $\rho_J = \rho$), and then derives their form for c_{J+1} under the assumption that retirees face an infinite horizon value function problem. In the data this assumption will fail, and so their parametrization of c_{J+1} will lead to different estimates from the parametrization of V_{J+1} . Because the parametrizations of V_{J+1} and C_{J+1} are not compatible with each other, their solution and that presented here are making different assumptions about retirement (retirement is unmodelled but implicit in the period $J + 1$ parameterization). If both approaches used a non-parametric formulation for period $J + 1$ we would expect them to give the same solution (Jorgensen and To, 2020). But as they differ in the parametrization of period $J + 1$ it is unsurprising we get slightly different results. When estimating using the V_{J+1} , we originally imposed that $\rho_J = \rho$, we got not dissimilar point estimates, but much larger standard deviations, and looking at J (the derivatives of the moments with respect to the estimated parameters) the derivatives with respect to ρ are negative, while those with respect to ρ_J are positive. Based on this we decided it was better to estimate separate values for ρ and ρ_J , and that this was more in line with the original estimates of Gourinchas and Parker (2002) in terms of the parameterization of the final period.

This example has served three purposes. First, it provides proof-of-concept implementation that GMM estimation of Life-Cycle models is feasible. Second, we saw in the previous section that the

³²See their equation (6) which states that $c_{J+1} = \gamma_0 + \gamma_1 x_{J+1}$ (both c_{J+1} and x_{J+1} are lower case as they renormalize the model prior to solving). By definition $\gamma_0 = h\gamma_1$, but the relation between κ and the gammas is more complex.

runtime of the steps in evaluating a parameter vector for GMM is faster than for SMM. Third, it demonstrates how the efficient GMM estimator is simpler and faster —because the moment conditions are separable— than would be the case for SMM, where two-iteration efficient SMM had to be used by Gourinchas and Parker (2002).

Conclusion

To motivate the use of Simulated Method of Moments estimation, the opening paragraph of McFadden (1989) states that “For some problems, the [model moments] may be difficult to express analytically or to compute, but relatively easy to simulate”. Developments over the last decade in both hardware and software have flipped this statement on its head for many Life-Cycle models, the moments of which can be simulated but are relatively easy to compute. Since SMM was always theoretically second best, in the sense that the asymptotic variance is a factor of $1 + 1/s$ times larger relative to GMM, the fact that for many Life-Cycle models GMM estimation is not only feasible but also faster suggests that SMM estimation is no longer a good idea. Simulated Method of Moments estimation will remain the only option for larger Life-Cycle models with higher-dimensional state spaces, but Generalized Method of Moments can be expected to become an appealing option for a growing range of Life-Cycle models.

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A Model Details

We described the model of Gourinchas and Parker (2002) in Section 4. This appendix just reports the values of all the pre-calibrated parameters which are set following Gourinchas and Parker (2002), and brief details of the computation.

Codes implementing the estimation (and the run-time results) can be found at:
<https://github.com/robertdkirkby/GourinchasParker2002>

Table A contains the calibrated values of the parameters of the model of Gourinchas and Parker (2002).

The parameters not mentioned in the table (other than those which will be estimated) are v and g . GP2002 set v , the age-dependent preference shifters, by first creating v_a as a fifth-order polynomial of age, and we get the polynomial coefficients from the code of Jorgensen (2023), and then some transformations of this get us to v itself: $v_a = 0 + 0.13964975 * age - 0.0047742190 * age^2 + 8.5155210 * 10^{-5} * age^3 - 7.9110880 * 10^{-7} * age^4 + 2.9789550 * 10^{-9} * age^5$. Then $v_{b,j} = \exp(\Delta_j v_{a,j})$, and finally set $v_1 = 1$ and iterate $v_j = v_{b,j} v_{j-1}^{1/(1-\rho)}$, for $j = 2, \dots, J$. Note that $v_{b,j} = (v_{j+1}/v_j)^{1/\rho}$ is all that is needed when solving the model using Euler equations as was done by previous studies (we have to use v , rather than v_b to be able to perform the value function iteration).

GP2002 set g , the age-dependent deterministic growth in earnings, based on the log-change in earnings as a fifth-order polynomial of age; the $\log(\text{earnings}) = 6.8013936 + 0.3264338 * age - 0.0148947 * age^2 + 0.000363424 * age^3 - 4.411685 * 10^{-6} * age^4 + 2.056916 * 10^{-8} * age^5$, and $g = \Delta_j \log(\text{earnings}_j)$.

Our initial values for the estimated parameters are $\beta = 0.9574$, $\rho = 0.6526$, $h = 0.0211$, $\kappa = 3$ (the first three are the estimates of Andrews et al. (2017), with $h = \gamma_0/\gamma_1$). Our initial value for ρ_J was set equal to the initial value for ρ .

Parameter	Value	Description
R	1.0344	Asset Returns
$\sigma_{z,n}$	$\sqrt{0.0212}$	Std. dev. of innovations to z
p	0.00302	Probability of zero earnings
σ_u	$\sqrt{0.0440}$	Std. dev. of e , when non-zero

Table 2: Calibrated Parameters of Gourinchas & Parker (2002)

The model is computed using VFI Toolkit. So the value function iteration is done with pure discretization, sped up with divide-and-conquer. The iteration of the agent distribution is done using the Tan improvement, as emphasized in the body of the paper. The optimization routine to minimize the GMM objective function is CMA-ES (Covariance Matrix Adaptation-Evolutionary Strategy), which is chosen as it is highly robust to non-linearities and local minima, although does

have slower runtimes than possible alternative algorithms. We discretize the i.i.d shocks using the an extension of the Tanaka-Toda method to handle age-dependent markov processes with drift. Note that we do not renormalize the problem to eliminate the unit-root markov shocks from the state space, which would be the cleaner/better/smarter approach but this model is small enough that we can afford to simply take a brute-force approach. We use 12 points to discretize the i.i.d. shocks, and 75 points to discretize the markov shocks; both are done with $nSigmas = 2$ (so plus/minus two standard deviations as the max/min grid points).

We recomputed the Jacobian matrix that determines the covariance matrix of the estimated parameter vector using larger grids, to ensure that our grids were large enough that numerical inaccuracy in the derivatives was not an issue.

Figure 1 shows the life-cycle profiles (age-conditional means) of consumption and income, both in the data, and for our estimates (the robust and efficient estimates have identical income processes, so only one is shown here), as well as the estimates for consumption of Andrews, Gentzkow, and Shapiro (2017).³³ We computed the GMM objective function based on the consumption moments reported by Andrews, Gentzkow, and Shapiro (2017) and it is smaller than that based on the consumption moments associated with our robust estimates, however it appears to be impossible to do as well as the AGS2017 estimates for any parametrization of the model we are estimating (because of the differences in the parametrization of the final period).

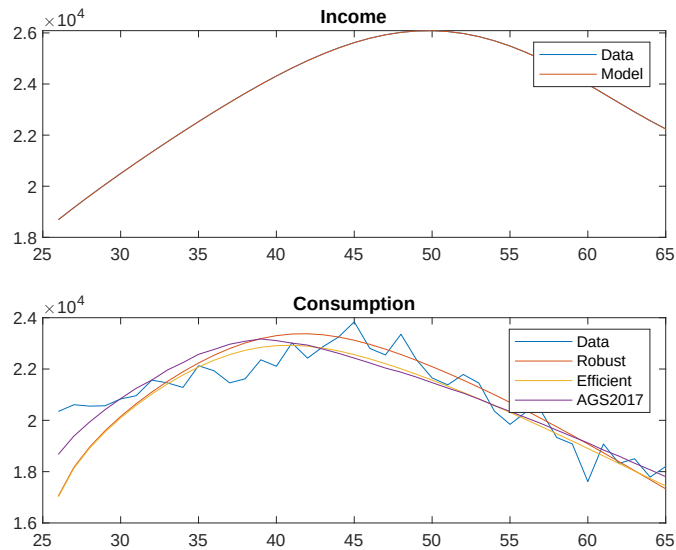


Figure 1: Age-conditional means of Income and Consumption in Data and Estimated Models

³³ Andrews, Gentzkow, and Shapiro (2017) have similar point estimates for the estimated parameters to Gourinchas and Parker (2002), and the mean consumption profile of the former is available, while the later is not available to be best of our knowledge.